over a calibrated scale and indicates the magnitude of electrical quantity.

Example: Ammeter, voltmeter and wattmeter.

ii. Integrating instruments

The instruments which measure the total quantity of electricity delivered over a specified time are called as integrating instruments.

Example: Energy meters (KWH meters) and ampere-hour meters

iii. Recording instruments

Instruments which give a continuous record of the variations of an electrical quantity over a period of time are called as recording instruments. The variations of the quantity being measured are recorded by a pen on a sheet of paper put over a moving drum.

Example: Electrocardiogram machine, pressure and temperature recorders.

6.2.3 Classification of measuring instruments according to the quantity being measured

Sl. No.	Name of the instrument	Quantity being measured
1	Voltmeter	Voltage
2	Ammeter	Current
3	Ohm meter	Resistance
4	Wattmeter	Power
5	Watt-hour meter	Energy
6	Power factor meter	Power Factor
7	Frequency meter	Frequency

6.3 Torques in indicating instruments

The following three types of torques are essential for the satisfactory operation of indicating instruments.

i. Deflecting torque
ii. Controlling torque and
iii. Damping torque

6.3.1 Deflecting torque

This torque is required in an indicating instrument for moving the pointer from its zero position. Deflecting torque is produced by utilizing the effects of electric current.

6.3.2 Controlling torque

This torque is required in an indicating instrument to ensure that the deflection is proportional to the magnitude of the quantity being measured. The controlling system produces a torque equal and opposite to the deflecting torque in order to make the deflection of pointer at a definite magnitude. The controlling system brings the moving system back to zero when the torque causing the deflection is removed.

(A) Controlling devices

There are two types of controlling devices.

i. Spring control
ii. Gravity control

i. Spring control

Figure 6.1 shows a commonly used spring control arrangement. Two hair springs made of phosphor bronze are attached to the moving system which

BEE_ENG_Chapter_06.indd 88 12/27/2019 10:30:42 AM